SetupX: Can LLM Agents Learn from Past Failures?

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arXiv 2605.26186v2 · Functionality-Correct Code Repository Setup

HarnessRSI 64 ★★

evidence: retained

Paper

Code

1 What changes, what is kept

1 Distill past setup trajectories
Commands + errors + verdicts when available.
Store root-cause advice, signals, executable atoms and hit / success / failure telemetry.
Even guilty runs can yield reusable fixes.

▼

2 Keep knowledge, not model weights
Semantic duplicate → merge signals + advice.
Other candidates → insert into the XPU store.
No held-out performance gate for admission is specified. Retrieval uses historical utility.

▼

3 Retrieve → snapshot → trial a repair
LLM picks up to 3 of 10 retrieved candidates.
Adapt commands; compare before / after errors.
Successful trial → keep container changes; otherwise restore. Can roll back multiple steps.

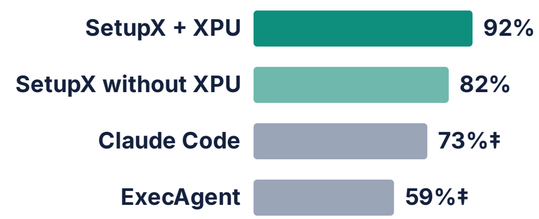
▼

4 Audit delayed outcomes
Next retrieval audits up to 5 intervening steps.
Adopted + resolved / improved → success.
Adopted + persistent / worse → failure.
Uncertain → neutral. Update ranking telemetry.

VERIFY → FINISH → final audit
Read-only in-loop verifier diagnoses errors.
Prosecutor probes; Judge checks each charge.
Fail: adverse verdict / timeout / no data.
Independent roles; same backbone model.

Original diagram · §§3–5, Appendices A–C. Local repair acceptance is not XPU admission.

2 92% pass; XPU adds 10 points



100 curated Python repos · qwen3.5-plus · fresh Ubuntu · 3600 s/run · common final audit.

Single-run fractions; no variance / CIs. Table 4 failures include missing evaluation data; cells do not identify it.

4 Where the gain is absent / uneven

- **Medium-difficulty gain:** 89%‡ vs 69%‡ without XPU. Low: 100%‡ vs 95%‡; high: 79%‡ vs 72%‡.
- **Reversals:** trax, neuralforecast, ajenti, civet pass without XPU, fail with it (Table 4). Cause unknown.
- **Multi-repo:** both ≥Mostly on 17/22; Full: SetupX 6, Qwen 1 (Table 2).
- **Service-isolation caveat:** stale Saleor occupies port 80 in Qwen's MISP run; no clean rerun reported (pp17–18).
- **Separate wording issue:** Appendix E's "four Partial/Shallow" conflicts with Table 2; we use the table.

‡ = PDF figure-label read; see SOURCE.md. No paper images shown.

3 Controls & remaining confounds

Retrieval ablation · same 100 repositories

Setting	Pass
LLM selector + clean KB	92%‡
Direct vector + clean KB	86%‡
LLM selector + noisy KB	88%‡
Direct vector + noisy KB	81%‡

Figure 6b · synthetic-noise stress test; no SD / CI. Opus 4.6: same 92%, API cost >10× (§6.4).

- **Identity exclusion:** 600 retained advice entries; target-repository entries removed. No related-repository leakage audit or cross-target reset protocol reported.
- **Shared conditions, unequal mechanisms:** Claude Code, OpenCode, Qwen Code, ExecAgent, Repo2Run and EnvBench each get the common model / timeout / final audit; no comparable XPU store reported for any.
- **Rollback is not unique:** Repo2Run has command-level rollback; SetupX has multi-step snapshot rollback. No rollback-only or telemetry-only ablation.
- **Selection / budgets:** tier defaults use current telemetry; no independent tuning split or baseline tuning budgets specified. Equal timeout ≠ equal retries / tokens.
- **Acquisition cost unknown:** no itemized donor runs, distillation, embedding, audit or snapshot cost. Long-term self-curation not isolated.

Takeaway

Retained repairs help: 92% vs 82% without XPU. Retrieval quality matters, but continuous self-improvement is not yet isolated or cost-accounted.

Our read · HarnessRSI

Score 64 · ★★ · evidence: retained

Reports: repo exclusion, ablations, shared timeout.

Not reported: variance, matched total compute, equal-memory baselines, acquisition cost.